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On averaged control for ergodic diffusions

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Talk at SAIFM 2026 conference

(partially joint with J. Djordjević, Uni of Niš, Serbia
and E.Yu. Yashchenko (Iashchenko), MSU)

Thanks and object of study

Congratulations to Albert Nikolaevich Shiryaev with his birthday!

Thanks the organisers for the invitation!

The object of study in this talk is the **ergodic** controlled SDE in dimension 1

$$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t, \quad X_0^\alpha = x, \quad (1)$$

with the cost to be minimized over Markov strategies,

$$\rho^\alpha(x) := \limsup_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{E}_x f(\alpha(X_t^\alpha), X_t^\alpha)\,dt.$$

Abstract

The ergodic Bellman equation is proved for a $d = 1$ controlled diffusion with variable diffusion and drift coefficients both depending on control; convergence of the Howard algorithm is established.

The **example** will relate to the controlled diffusion with additive Wiener process and may be regarded as a toy ergodic version of the results by M. Taksar, see the 2nd ref.


S.V. Anulova, H. Mai, A.Yu. Veretennikov, On iteration improvement for averaged expected cost control for one-dimensional ergodic diffusions, SIAM J. Control Optim. 2020, 58(4), 2312-2331.


Taksar, M. Optimal risk and dividend distribution control models for an insurance company. Math. Methods of OR 51, 1-42 (2000). https://doi.org/10.1007/s001860050001

References

The classical reference is the book by Araposthatis, Borkar and Ghosh 2012. Basically, they treat the case $\sigma(x)$ not depending on control

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Here are some my further recent papers on the topic.

A.Yu.Veretennikov, On averaged expected cost control for 1D controlled ergodic diffusions with switching, MPRF, 2023, 23(2), 259 - 294.

A.Yu. Veretennikov, On Averaged Control and Iteration Improvement for a Class of Multidimensional Ergodic Diffusions, in: Kolmogorov Operators and Their Applications, S. Menozzi (ed.), A. Pascucci (ed.), S. Polidoro (ed.), ser. CONFERENCE PROCEEDINGS, Springer Singapore (Singapore), 2024, 315-349. https://doi.org/10.1007/978-981-97-0225-1_10

A.Yu. Veretennikov, E.Yu. Iashchenko, A primer of optimal ergodic average control in a diffusion model of an insurance company (to appear), Automat. and Rem. Control, 2026, no. 12, 97-114 (in Russian).

The setting $dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t \quad (1)$

Let a non-empty compact set $U \subset \mathbb{R}^\ell$ be a range of possible control values. Let $b : U \times \mathbb{R}^1 \to \mathbb{R}$, $\sigma : U \times \mathbb{R}^1 \to \mathbb{R}$, $\alpha : \mathbb{R}^1 \to U$ be given Borel functions; some more regularity assumptions will be presented later on.
Denote by L^α the generator, which corresponds to the equation (1) with a fixed function $\alpha(\cdot)$ depending on x:

$$L^\alpha h(x) = b(\alpha(x), x)h'(x) + \frac{\sigma^2(\alpha(x), x)}{2}h''(x), \ x \in \mathbb{R}^1. \quad (2)$$

Note that according to [Krylov, On the selection of a Markov process... 1973] the equation (1) has a weak solution which is a homogeneous Markov process. It is known that in dimension one it is weakly unique. For the existence of a strong solution we can mollify the strategy keeping the discrepancy $\|L^\alpha v(x) - L^{\tilde\alpha} v(x)\|_{L_1}$ small for specially chosen sequence of auxiliary functions v, see what follows.

The setting: averaged cost

$$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t \qquad (1)$$

Given a **running cost function** $f : U \times \mathbb{R} \to \mathbb{R}^1$ from a suitable function class $\mathcal{K}$ – which will be defined shortly – our goal is to choose an optimal, or nearly-optimal stationary Markov control strategy $\alpha : \mathbb{R}^1 \to U$ such that the corresponding solution X^α minimizes the averaged cost function

$$\rho^\alpha(x) := \limsup_{T\to\infty} \frac{1}{T} \int_0^T \mathsf{E}_x f(\alpha(X_t^\alpha), X_t^\alpha)\,dt. \qquad (3)$$

The class of such strategies with a strong solution of the SDE (1) will be denoted by $\mathcal{A}$; they will be called admissible. However, weak solutions are also useful here.

Strong solutions are preferred and can be tackled; but then the RIA and the proofs will be a bit more involved. Strong solutions are studied in our work in progress with J. Djordjević.

The setting

$$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t \qquad (1)$$

Suppose that for every $\alpha \in \mathcal{A}$ the solution X^α of the equation (1) is a homogeneous Markov ergodic process, for which there exists a limiting and invariant distribution μ^α of the process (X_t^α), $t \to \infty$, the same for all initial conditions $X_0 = x \in \mathbb{R}^d$,

$$\rho^\alpha(x) \equiv \rho^\alpha := \int f^\alpha(x)\,\mu^\alpha(dx) =: \langle f^\alpha, \mu^\alpha \rangle, \qquad (4)$$

and

$$\rho(x) \equiv \rho := \inf_{\alpha \in \mathcal{A}} \int f^\alpha(x)\,\mu^\alpha(dx) = \inf_{\alpha \in K} \langle f^\alpha, \mu^\alpha \rangle. \qquad (5)$$

The goal is to find the optimal cost ρ and the strategy α on which this value is attained or nearly attained. Recall that strong solutions are preferred.

Auxiliary function $v^\alpha(x) := \int_0^\infty \mathsf{E}_x(f^\alpha(X_t^\alpha) - \rho^\alpha)\,dt$

$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t; \quad \rho^\alpha = \langle f^\alpha, \mu^\alpha \rangle$

Note that under our assumptions below on the growth bounds of f and on the ergodicity properties of the process with any strategy, the value of ρ will not depend on x. Let us define auxiliary functions

$$v^\alpha(x) := \int_0^\infty \mathsf{E}_x(f^\alpha(X_t^\alpha) - \rho^\alpha)\,dt, \qquad \alpha \in \mathcal{A}. \tag{6}$$

Under the ergodicity assumptions below the integral in (6) will converge.

NB: in the ergodic setting other strategies different from "Markov" or "stationary" ones are not considered.

First goal: ergodic HJB equation

$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t; \quad \rho^\alpha = \langle f^\alpha, \mu^\alpha \rangle$

The first goal of this paper is to prove that the cost ρ – which is a constant in the ergodic setting – is the component of the pair (V, ρ), which is a unique in the appropriate class solution of the *ergodic Bellman's equation,*

$$\boxed{\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0, \quad x \in \mathbb{R},} \tag{7}$$

where similarly to (2), L^u for $u \in U$ is defined by the formula

$$L^u f(x) := b(u,x)f'(x) + \frac{1}{2}\sigma^2(u,x)f''(x), \qquad x \in \mathbb{R}, \tag{8}$$

and that the equation (7) has a unique solution in a certain class of functions. For the setting with weak solutions it is known [Anulova, Mai, Veretennikov 2020] and allows to find an optimal strategy. However, in the strong solution setting such a strategy may be inadmissible and must be mollified.

Second goal: RIA convergence

Reward Improvement Algorithm

The second goal is to show convergence of the Howard's "reward improvement algorithm", or, "RIA", or, in some papers, "PIA" (for "policy improvement algorithm") to the costs, ρ, see the details of this algorithm below.

For ergodicity, we will assume recurrence conditions, which provide a uniform recurrence for *any* strategy (so called *"blanket recurrence conditions"*). It is likely that such restrictions may be relaxed to the "near monotonicity" type conditions as in



A. Arapostathis, V.S. Borkar, M.K. Ghosh. *Ergodic control of diffusion processes.* Cambridge: Cambridge University Press, 2012.

Yet to the best of our knowledge, to this moment it is still an open problem.

We assume the following

$dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t$

(A0) $\sigma(u,x) \in C^1$ in u uniformly wrt x.

(A1) The drift b: $|b(u,x)| \le C_b$, and (may be relaxed)

$$\lim_{|x|\to\infty} \sup_{u\in U} \langle x\, b(u,x)\rangle = -\infty. \tag{9}$$

(A2) The function σ is bounded, $|\sigma(u,x)| \le C_\sigma$, uniformly non-degenerate, $|\sigma(u,x)|^{-1} \le C_\sigma$.

(A3) The function f belongs to the class $\mathcal{K}$ of functions which admit a uniform in u polynomial bound: there exist constants $C_1, m_1 > 0$ such that for any x,

$$\sup_{u\in U} |f^u(x)| \le C_1(1 + |x|^{m_1}).$$

(A4) Functions $\sigma(u,x), b(u,x), f^u(x)$ are continuous in u, $\forall x$.

(A5) The set U is compact.

(A6) $\sigma(u,x)$, $b(u,x)$, $f^u(x) \in C^1$ in x for every u.

Lemma 1

(1) $dX_t^\alpha = b(\alpha(X_t^\alpha), X_t^\alpha)\,dt + \sigma(\alpha(X_t^\alpha), X_t^\alpha)\,dW_t,\ X_0^\alpha = x$

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Recall that b and σ are assumed bounded and σ is non-degenerate.

Lemma (1)

1. Weak solution (X) for the equation (1) exists for any strategy α. It is unique in distribution and, hence, is Markov and strong Markov.

2. If in addition $\sigma(\cdot)$ is locally Lipschitz (or even Hölder $1/2$), then strong solution exists.

In D1 the claim 2 is well-known: [Zvonkin, 1974] even in the non-homogeneous case. See some recent "multidimensional" extension in the 2026 article www.mathnet.ru/rus/tvp5906/ (arXiv:2510.04329) (Eng.: https://doi.org/10.1137/S0040585X97T992938).

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Lemma (2)

Let the assumptions (A1) – (A3) hold true. Then

- *For any $C_1, \ell_1 > 0$ there exist $C_2, \ell_2 > 0$ such that for any strategy $\alpha \in \mathcal{A}$ and for any function g growing no faster than $C_1(1 + |x|^{\ell_1})$ uniformly with respect to z,*

$$\sup_t |\mathsf{E}_x g(X_t^\alpha)| \le C(1 + |x|^{\ell_2}). \tag{10}$$

- *For any $\alpha \in \mathcal{A}$, the invariant measure μ^α integrates any polynomial and, moreover,*

$$\sup_{\alpha\in\mathcal{A}} \int |x|^k\, \mu^\alpha(dx) < \infty, \quad \forall\, k > 0. \tag{11}$$

- *For any $\alpha \in \mathcal{A}$ the function $\rho^\alpha = const\,(\alpha) \le C < \infty$.*

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Lemma (2, ctd)

- *ρ^α is uniquely determined for any strategy α.*
- *Moreover, for any $k > 0$ and $f \in \mathcal{K}$,*

$$\sup_{\alpha\in\mathcal{A}} t^k \left|\mathsf{E}_{x,z} f^\alpha(X_t^\alpha) - \rho^\alpha\right| \to 0, \quad t\to\infty, \tag{12}$$

and

$$\sup_{\alpha\in\mathcal{A}} \left| \frac{1}{T}\int_0^T \mathsf{E}_x f^\alpha(X_t^\alpha)\, dt - \rho^\alpha \right| \to 0, \quad T\to\infty. \tag{13}$$

Lemma 3 $L^\alpha h(x) = b(\alpha(x), x)h'(x) + \frac{\sigma^2(\alpha(x),x)}{2}h''(x)$

Recall that $\langle f^\alpha, \mu^\alpha \rangle = \rho^\alpha$

Lemma (3)

Let (A1) – (A4) be satisfied. Then

- *The function v^α is uniquely determined by eq-n (6); $v^\alpha \in W^2_{p,loc}$ for any $p \geq 1$ and satisfies in the Sobolev sense a Poisson equation in the whole space,*

$$L^\alpha v^\alpha + f^\alpha - \langle f^\alpha, \mu^\alpha \rangle = 0. \tag{14}$$

In particular, for almost every $x \in \mathbb{R}^d$

$$L^\alpha(x) v^\alpha(x) + f^\alpha(x) - \langle f^\alpha, \mu^\alpha \rangle = 0, \tag{15}$$

or, equivalently,

$$\frac{1}{\sigma^2(\alpha(x), x)} L^\alpha(x) V(x) + \frac{f^\alpha(x)(x)}{\sigma^2(\alpha(x), x)} - \frac{\rho^\alpha}{\sigma^2(\alpha(x), x)} = 0. \tag{16}$$

Lemma 3, ctd

$L^\alpha h(x) = b(\alpha(x), x)h'(x) + \frac{\sigma^2(\alpha(x),x)}{2} h''(x)$

Recall that $\langle f^\alpha, \mu^\alpha \rangle = \rho^\alpha$

Lemma (3, ctd)

- *Solution of the equation (14) is unique up to an additive constant in the class of Sobolev solutions $W^2_{p,loc}$ in x with a no more than some (any) polynomial growth.*
- $\langle v^\alpha, \mu^\alpha \rangle = \int v^\alpha(x)\mu^\alpha(dx) = 0.$
- *For any strategy α the function v^α is continuous as well as $(v^\alpha)'_x$, and there exist $C, m > 0$ both depending only on the constants in (A1)–(A3) such that*

$$\sup_\alpha (|v^\alpha(x)| + |v^\alpha(x)'_x|) \le C(1 + |x|^\ell). \qquad (17)$$

- $v^\alpha \in C^{1,Lip}$ *in x (i.e., $(v^\alpha)'_x$ is locally Lipschitz).*

Lemma 4

$\lim_{|x|\to\infty} \sup_{u\in U} \langle x\, b(u,x)\rangle = -\infty$; σ^2 non-degenerate; b, σ bounded

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Lemma (4)

Let the assumptions (A1) – (A2) hold true. Then

$$\Lambda \ll \bar{\mu}^{\alpha} \ll \Lambda$$

($\Lambda \sim \mu^{\alpha}$), where Λ is the Lebesgue's measure in $\mathbb{R}^1$.
Moreover, *if all functions $g \geq 0$ from some family possess a uniform polynomial bound $g(x) \leq C(1 + |x|^{\ell})$ and the value $\langle g, \bar{\mu}^{\alpha}\rangle$ is uniformly small, then for any $R > 0$ the L_1-norm $\|1(|\cdot| \leq R)g(\cdot)\|_{L_1}$ is uniformly small. More precisely, for any $\delta > 0$ and any $R > 0$ there exists $\epsilon > 0$ such that if $\langle g, \bar{\mu}^{\alpha}\rangle < \epsilon$, then*

$$\|1(|\cdot| \leq R)\, g(\cdot)\|_{L_1} \leq \delta.$$

NB: In 1D $\exists$ an explicit formula for the invariant density.

Comment on lemma 4 $\Lambda \ll \bar{\mu}^{\alpha}$

Why it is needed? For the applicability of Krylov's inequalities.

NB: In the proof of the Theorem below (truncated) it will be used in the following form: for any $\delta > 0$ there exists $\epsilon > 0$ such that

$$\psi_n^2 : \ \langle \psi_n^2, \mu^{\alpha} \rangle \leq \epsilon$$

implies that there exists $n(\delta)$ such that

$$\sup_{n \geq n(\epsilon)} \int_{|x| \leq N} \psi_n^2(x)\, dx < \delta.$$

This is what is required for using one of Krylov's inequalities.

The proof follows easily from the explicit formula for the invariant density in $d = 1$.

RIA (Reward Improvement (Howard – Bellman's) Algorithm)

(HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$; the cost "function" (constant) is ρ

The exact RIA which provides a sequence approximating ρ reads as follows. We start with some homogeneous Markov strategy α_0, which uniquely determines $\rho_0 = \rho^{\alpha_0} \equiv \langle f^{\alpha_0}, \mu^{\alpha_0}\rangle$ and $v_0 = v^{\alpha_0}$. Next, for any couple (v,ρ) : $v \in \bigcap_{p>1} W^2_{p,loc}$, $\rho \in \mathbb{R}$, define

$$F[v,\rho](x) := \inf_{u\in U}[L^u v(x) + f^u(x) - \rho]\,.$$

By induction, given α_n, ρ_n and v_n, the next "improved" strategy α_{n+1} is defined as follows: for a.e. x the (Borel measurable) function α_{n+1} is chosen so that

$$\begin{aligned} L^{\alpha_{n+1}} v_n(x) + f^{\alpha_{n+1}}(x) - \rho_n &= F[v_n,\rho_n](x,z) \\ &\equiv \inf_{u\in U}[L^u v_n(x) + f^u(x) - \rho_n]\,. \end{aligned} \tag{18}$$

RIA

(HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$; the cost "function" (constant) is ρ

Note that

$$L^{\alpha_{n+1}} v_n(x) + f^{\alpha_{n+1}}(x) - \rho_n = F[v_n, \rho_n](x)$$

$$\equiv \inf_{u\in U} \left[L^u v_n(x) + f^u(x) - \rho_n\right],$$

is equivalent to

$$L^{\alpha_{n+1}} v_n(x) + f^{\alpha_{n+1}}(x) = \inf_{u\in U} \left[L^u v_n(x) + f^u(x)\right],$$

hence, define $G[v_n](x) := \inf_{u\in U} \left[L^u v_n(x) + f^u(x)\right]$, and also to

$$\alpha_{n+1}(x) \in \mathrm{Arginf}_{u\in U} \left[L^u v_n(x) + f^u(x)\right].$$

This strategy will be further smoothed, see what follows.

RIA

(HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$; the cost "function" (constant) is ρ

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We assume that a Borel measurable version of such strategy may be chosen: use measurable choice theorems, e.g., by Shchegolkov 1948. The value ρ_{n+1} then could be defined as

$$\rho_{n+1} := \langle f^{\alpha_{n+1}}, \mu^{\alpha_{n+1}} \rangle,$$

where in turn, $\mu^{\alpha_{n+1}}$ is the (unique) invariant measure, which corresponds to the strategy α_{n+1}. The same is applied if this strategy is smoothed. Also, recall that

$$v_n(x) = \int_0^\infty \mathrm{E}_x(f^{\alpha_n}(X_t^{\alpha_n}) - \rho_n)\, dt,$$

and

$$v_{n+1}(x) = \int_0^\infty \mathrm{E}_x(f^{\alpha_{n+1}}(X_t^{\alpha_{n+1}}) - \rho_{n+1})\, dt.$$

However, strong SDE solution may not exist is α_{n+1} is not smooth enough! So, we need to mollify it.

RIA2 (for strong solutions)

(HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$; the cost "function" (constant) is ρ

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The approximate RIA2 which provides a sequence approximating ρ reads as follows. Again we start with some homogeneous Markov strategy α_0, which uniquely determines $\rho_0 = \rho^{\alpha_0} \equiv \langle f^{\alpha_0}, \mu^{\alpha_0}\rangle$ and $v_0 = v^{\alpha_0}$. Next, for any couple (v,ρ) : $v \in \bigcap_{p>1} W^2_{p,loc}$, $\rho \in \mathbb{R}$, define

$$F[v,\rho](x) := \inf_{u\in U}[L^u v(x) + f^u(x) - \rho]\,.$$

By induction, given α_n, ρ_n and v_n, the next "improved" strategy α_{n+1} is defined as follows: for a.e. x the function α_{n+1} is chosen so that it is Lipschitz in x and

$$\begin{aligned} L^{\alpha_{n+1}} v_n(x) + f^{\alpha_{n+1}}(x) - \rho_n &\le F[v_n,\rho_n](x) + h_n(x) \\ &\equiv \inf_{u\in U}[L^u v_n(x) + f^u(x) - \rho_n] + h_n(x), \end{aligned} \tag{19}$$

with $\|h_n\|_{L_1(\mathbb{R})} \le \frac{N\delta}{(n+1)(n+2)}$, & $\delta \in (0,1)$ is chosen arbitrarily.

Theorem 1 (HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$

Using weak SDE solutions

Use RIA-1 and weak SDE solutions.

Theorem (1)

Let the assumptions (A1) – (A5) be satisfied. Then:
1. For any n, $\rho_{n+1} \le \rho_n$, and there is a limit $\rho_n \downarrow \tilde{\rho}$.
2. The sequence (v_n) is pre-compact in $C^1[-N, N]$ in x for each $N > 0$.
3. If $\tilde{v}$ is any limiting point of the family $(v_n, n \ge 1)$ in C^1_{loc} in the variable x, then the couple $(\tilde{v}, \tilde{\rho})$ solves the equation (7).
4. This solution $(\tilde{v}, \tilde{\rho})$ is unique in the class of functions growing no faster than some polynomial and belonging to the class $W^2_{p,loc}$ in x for any $p > 0$ and any z.
5. The component $\tilde{\rho}$ in the couple $(\tilde{v}, \tilde{\rho})$ coincides with ρ.
6. In addition, if (A6) is satisfied, then $\tilde{v}'' \in Lip_{loc}$ in x for each z.

Theorem 2 (HJB) $\inf_{u\in U}[L^u V(x) + f^u(x) - \rho] = 0$

Using strong SDE solutions

Use RIA-2 and strong SDE solutions.

Theorem (2)

Let the assumptions (A0) – (A5) be satisfied. Then:
1. For any n, $\rho_{n+1} + \frac{N\delta}{(n+2)} \le \rho_n + \frac{N\delta}{(n+1)}$*, and there is a limit* $\rho_n \to \tilde{\rho}$*; here (an arbitrary)* $\delta \in (0,1)$ *is from (19).*
2. The sequence (v_n) *is pre-compact in* $C^1[-N,N]$ *in x for each* $N > 0$*.*
3. If $\tilde{v}$ *is any limiting point of the family* $(v_n, n \ge 1)$ *in* C^1_{loc} *in the variable x, then the couple* $(\tilde{v}, \tilde{\rho})$ *solves the equation (7).*
4. This solution $(\tilde{v}, \tilde{\rho})$ *is unique in the class of functions growing no faster than some polynomial and belonging to the class* $W^2_{p,loc}$ *in x for any* $p > 0$ *and any z.*
5. The component $\tilde{\rho}$ *in the couple* $(\tilde{v}, \tilde{\rho})$ *coincides with* ρ*.*
6. In addition, if (A6) is satisfied, then $\tilde{v}'' \in Lip_{loc}$ *in x for each z.*

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A primer with $\sigma \equiv 1$

Based on the paper with Elizaveta Iashchenko (to appear: AiT 2026 no.12)

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Here is an ergodic analogue of the controlled diffusion model in the insurance company risk analysis by M. Taksar.

Let

$$\boxed{dX_t = (\mu - a(X_t))dt + dW_t, \quad X_0 = x > 0, \quad 0 \le a(\cdot) \le M,} \tag{20}$$

under the assumption that

$$0 < \mu < M.$$

Here $a(\cdot)$ is the (ergodic) strategy to be chosen so as to satisfy

$$a(x) \equiv 0, \quad \forall\, x \le 0,$$

so as to maximize the average limit

$$\liminf_{T\to\infty} \frac{1}{T} \mathsf{E}_x^{a(\cdot)} \int_0^T a(X_t)\, dt \to \sup_{a(\cdot)} =: r. \tag{21}$$

Ergodic strategies. The hypothesis.

Notice that all Markov strategies are admissible due to Zvonkin (1974)

One evident ergodic strategy here is

$$a(x) = M\,1(x > x_0), \qquad \text{with some } x_0 \geq 0.$$

The hypothesis: the average cost function (in fact, a constant) is

$$\boxed{r = \mu.} \tag{22}$$

Definition

The strategy $a(\cdot)$ as well as the related SDE solution X are called ergodic if there exists a unique invariant measure μ^a which does not depend on the initial value $X_0 = x$ such that fpor any bounded, Borel measurable function g and for any i.c. x,

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{E}_x^a g(X_t)dt = \int g(y)\mu^a(dy).$$

Theorems 3 and 4

No. 4 yet non-rigorous!

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Theorem (3)

The cost function (actually, the constant) r in this model equals

$$r = \mu.$$

This cost is attained on any ergodic strategy $a(\cdot)$ satisfying the condition

$$\lim_{t\to\infty} \frac{\mathrm{E}^a X_t}{t} = 0. \tag{23}$$

The proof follows from Ito's formula applied to X_t.

Theorem (4)

Ergodic Bellman equation (as a hypothesis):

$$\sup_{a\in[0,M]} \left[(\mu - a)V'(x) + \frac{1}{2}V''(x) + a - \rho\right] = 0, \tag{24}$$

Proof of theorem 3

Condition (23) $\mathsf{E}^a X_t/t \to 0$ as $t \to \infty$

For any ergodic strategy $a(\cdot)$ we have due to 20),

$$\mathsf{E}_x^a X_T - x = \mathsf{E}^a \int_0^T (\mu - a(X_s))ds.$$

Divide by t and let $t \to \infty$. By virtue of (23) the limit exists

$$\lim_{T\to\infty} T^{-1}\mathsf{E}^a \int_0^T (\mu - a(X_s))ds = 0.$$

So,

$$\lim_{T\to\infty} T^{-1}\mathsf{E}^a \int_0^T a(X_s)ds = \mu.$$

Hence,

$$r = \mu.$$

Clearly, this cost is attained on any ergodic strategy a. □

Explicit solution of Bellman equation in Thm 4

Notice that we did not use Bellman equation in the above proof

Use that $r = \mu$. Then the other component of solution of ergodic Bellman equation equals

$$V(x) = x + C.$$

This is easy to check and can be found by solving the second order ODEs on the left and on the right of the chosen value x_0, in which value the continuous glueing condition[1] can be used.

[1] This condition itself may be rigorously verified for this model.

Optimal strategy via ergodic Bellman equation in Thm 4

Once the pair (V, r) is found, the optimal strategy for the running cost $f(a, x)$ may be constructed in the form

$$\bar{a}(x) \in \operatorname*{Argsup}_a \left[(\mu - a) V'(x) + \frac{1}{2} V''(x) + f(a, x) - \rho \right], \tag{25}$$

where in our case $f(a, x) \equiv a$, so that, equivalently,

$$\bar{a}(x) \in \operatorname*{Argsup}_a \left[a - aV'(x) \right]$$

Since $V'(x) = 1$, we get that a can be any ergodic strategy satisfying the condition (23): $\mathrm{E}^a X_t / t \to 0$ as $t \to \infty$. As we see, this does not contradict the answer obtained previously without using ergodic Bellman equation.

THANKS FOR YOUR ATTENTION